

Soln: p: You can ride the roller coaster q: You are under 4 feet tall
r: You are older than 16 years old

$[(q \rightarrow \neg p) \vee (r \rightarrow p)] \rightarrow \text{My soln. ??}$

Book solution: $[(\neg q \wedge \neg r) \rightarrow \neg p]$

System Specifications

Translating natural languages (such as English) to logical expressions is an essential part of specifying both hardware and software systems. System and software engineers take requirements in natural language and produce precise and unambiguous specifications that can be used as the basis for system development. System specifications should be consistent, that is, they should not contain conflicting requirements that could be used to derive a contradiction. When specifications are not consistent, there would be no way to develop a system that satisfies all specifications.

Example 4: Determine whether these system specifications are consistent:

- "The diagnostic message is stored in the buffer or it is retransmitted."
- "The diagnostic message is not stored in the buffer."
- "If the diagnostic message is ~~not~~ stored in the buffer, then it is retransmitted."

Soln: p: The diagnostic message is stored in the buffer
q: The diagnostic message is retransmitted.

$(p \vee q)$ is True $\neg p$ is True $p \rightarrow \neg q$ is True
By defn. of negation, p is False. $\therefore p \rightarrow \neg q$ is True (consistent)
By defn. of OR, q is True. The specifications are consistent.

Logic Puzzles

Example 7: As a reward for saving his daughter from pirates, the King has given you the opportunity to win a treasure hidden inside one of three trunks. The two trunks that do not hold the treasure are empty. To win, you must select the correct trunk. Trunks 1 and 2 are each described.